LEZIONE 3

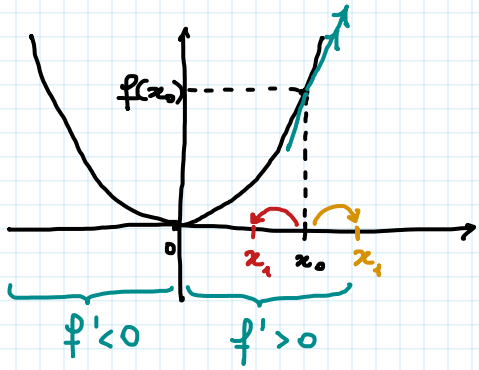
ALGORITMO di DISCESA del GRADIENTE

$f : \mathbb{R}^N \longrightarrow \mathbb{R}$ $\qquad \nabla f = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_N} \right)$

PROBLEMA: Trovare un MINIMO della funzione f

ESEMPIO per $N = 1$

$f : \mathbb{R} \longrightarrow \mathbb{R}$
$x \longmapsto x^2$

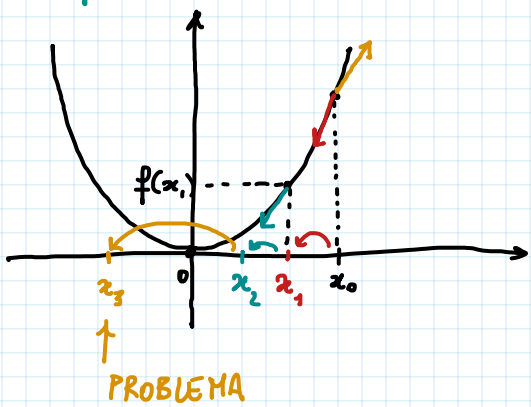


MINIMO in $x = 0$

$f'(x_0) = 2x_0 > 0$

→ f' mi porta nella direzione di massima crescita

→ DIREZIONE OPPOSTA : $-f'$



1) Scegliamo $x_0 \in \mathbb{R}$ nel DOMINIO di f

2) Ci spostiamo nel verso OPPOSTO rispetto a f', lungo la retta tg

$$x_1 = x_0 - \eta f'(x_0) \qquad \eta \in \mathbb{R}, \text{ IPERPARAMETRO}$$

3) Ripetiamo per m iterazioni

$$x_2 = x_1 - \eta f'(x_1) \qquad \dots \qquad x_m = x_{m-1} - \eta f'(x_{m-1})$$

$$x_3 = x_2 - \eta f'(x_2)$$

METODO ITERATIVO - DISCESA del GRADIENTE per $N=1$

$$\begin{cases} \boxed{x_0} = a \in \mathbb{R} \\ x_n = x_{n-1} - \eta f'(x_n) \end{cases}$$

dove $\eta \in \mathbb{R}$ sono IPERPARAMETRI
$n \in \mathbb{N}$

OBIETTIVO. $|x_n - m| < \varepsilon$

$|f(x_n) - f(m)| < \varepsilon$

$\boxed{|f'(x_n)| < \varepsilon} \longrightarrow$ Se $f'(x_n) = 0$ allora

$$x_{n+1} = x_n - \eta \cancel{f'(x_n)}^{\,0} = x_n$$

oppure $n > T$ $\rightsquigarrow$ TERMINAZIONE
T ↑ soglia

HW. $f: \mathbb{R} \to \mathbb{R}$
$x \mapsto x^2$

1) $x_0 = 1$, $\eta = \frac{1}{10}$

2) $x_0 = 1$, $\eta = 1$

Anche con Python / Numpy

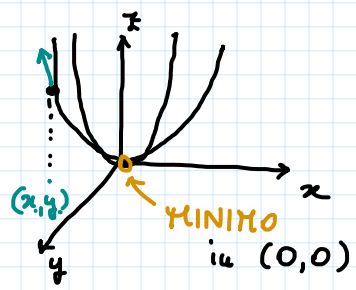
METODO ITERATIVO - DISCESA del GRADIENTE per $N > 1$

$f: \mathbb{R}^N \to \mathbb{R}$

$$\boxed{\begin{cases} \underline{x}_0 = \underline{a} \in \mathbb{R}^N \\ \underline{x}_n = \underline{x}_{n-1} - \eta \cdot \nabla f(\underline{x}_{n-1}) \end{cases} \qquad \eta \in \mathbb{R} \text{ IPERPARAMETRO}}$$

ESEMPIO $N=2$

$f: \mathbb{R}^2 \to \mathbb{R}$
$(x,y) \mapsto x^2 + y^2$



$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right) = (2x, 2y)$$

$\frac{\partial f}{\partial x}(x_0, y_0) = 2x_0 > 0$ $\qquad$ $\frac{\partial f}{\partial y}(x_0, y_0) = 2y_0 > 0$

STRATEGIE di TERMINAZIONE

$$\| \underline{x}_n - \underline{m} \| < \varepsilon$$

$$| f(\underline{x}_n) - f(\underline{m}) | < \varepsilon$$

$$\| \underline{x}_n - \underline{x}_{n-1} \| < \varepsilon$$

$$\| \nabla f(\underline{x}_n) \| < \varepsilon$$

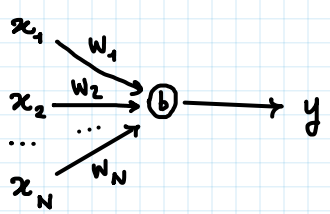
$$n > T$$

HW: Implementare l'algoritmo per $f(x,y) = x^2 + y^2$ per $\eta = \frac{1}{10}$, $\eta = \frac{1}{2}$, $\eta = 1 \dots$ e $(x_0, y_0) = (1,1)$

Python / Numpy

RETI NEURALI

MODELLO BASE: PERCETTRONE / NEURONE



$f : \mathbb{R}^N \to \mathbb{R}$

$f_{\underline{w},b}(\underline{x})$

$\underline{w}$, b sono PARAMETRI

$\underline{x} \in \mathbb{R}^N$ $\quad \underline{w} \in \mathbb{R}^N$ $\quad b \in \mathbb{R}$ $\quad y \in \mathbb{R}$

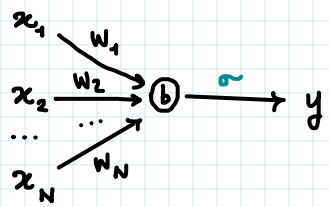
In particolare:

$$f(\underline{x}) = \begin{cases} 0 & \text{se } \underbrace{\sum_{i=1}^{N} x_i w_i}_{\underline{x} \cdot \underline{w}} \leq T \\ 1 & \text{se } \sum_{i=1}^{N} x_i w_i > T \end{cases} = \begin{cases} 0 & \underline{x}\,\underline{w} + b \leq 0 \\ 1 & \underline{x}\,\underline{w} + b > 0 \end{cases}$$

percettrone ATTIVATO

$b = -T$

Ciò che DETERMINA l'OUTPUT è $\boxed{\underline{x}\,\underline{w} + b}$ ≤ 0 NON ATTIVO; > 0 ATTIVO

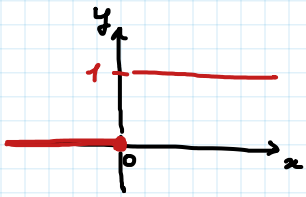


$$\boxed{f(\underline{x}) = \sigma(\underline{x}\cdot\underline{w} + b)}$$

FUNZIONE di ATTIVAZIONE

$\sigma : \mathbb{R} \to \mathbb{R}$

$$\sigma(x) = \begin{cases} 0 & \text{se } x \leq 0 \\ 1 & \text{se } x > 0 \end{cases}$$



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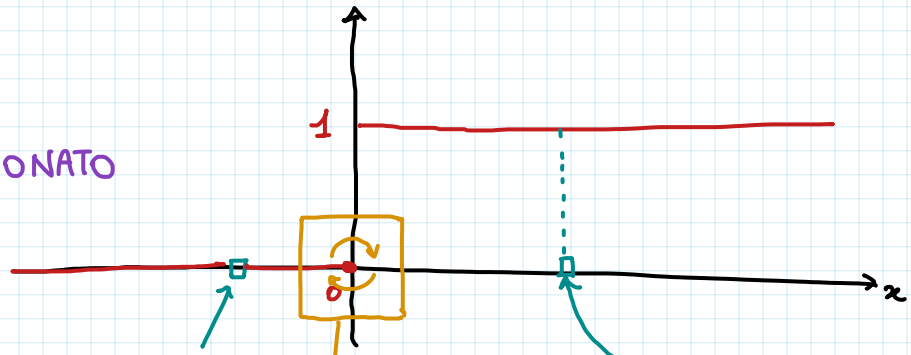
FUNZIONE di HEAVISIDE

IL PROBLEMA è MAL-CONDIZIONATO



$\sigma(\underline{x}\cdot\underline{w} + b)$ → $y = 0$

$\underline{x} + \Delta\underline{x}$
$\underline{w} + \Delta\underline{w}$
$b + \Delta b$

$\sigma((\underline{x}+\Delta\underline{x})\cdot(\underline{w}+\Delta\underline{w}) + b + \Delta b)$

SPOSTAMENTI PICCOLI dei PARAMETRI causano un CAMBIAMENTO COMPLETO

ALTRE FUNZIONI di ATTIVAZIONE

Funzioni $\sigma : \mathbb{R} \to \mathbb{R}$ NON decrescenti, PROPRIETÀ UTILI:

1) σ sia CONTINUA, $\sigma \in \mathcal{C}^0(\mathbb{R})$
2) σ sia CONTINUA, DERIVABILE e DERIVATE CONTINUE, $\sigma \in \mathcal{C}^1(\mathbb{R})$ 1 volta
   $\sigma \in \mathcal{C}^2(\mathbb{R})$ 2 volte
   ...
   $\sigma \in \mathcal{C}^\infty(\mathbb{R})$ ∞ volte
3) σ sia LIMITATA

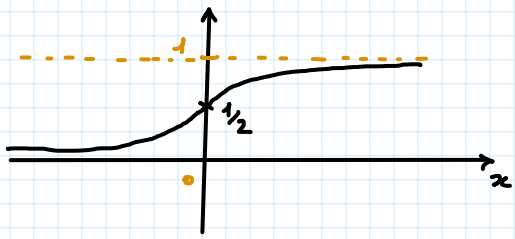
   $\exists L \in \mathbb{R}, L \geq 0 : \forall x \in \mathbb{R}, |\sigma(x)| \leq L$

⟶ PERCETTRONE con σ Fz di ATTIVAZIONE almeno 1

⟶ Si dice NEURONE ARTIFICIALE

1) SIGMOIDE LOGISTICA

$\sigma : \mathbb{R} \to \mathbb{R}$ $\quad \sigma \nearrow$ $\quad \sigma \in \mathcal{C}^\infty(\mathbb{R})$ $\quad |\sigma| \leq 1$



$\beta \in \mathbb{R}$ PARAMETRO

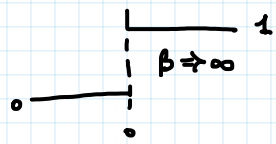
$\beta > 0$

$$\sigma_\beta(x) = \frac{1}{1+e^{-\beta x}}$$

$$\frac{d\sigma}{dx}(x) = \frac{\beta e^{-\beta x}}{(1+e^{-\beta x})^2} = \beta \frac{1}{1+e^{-\beta x}} \cdot \frac{\overbrace{e^{-\beta x}}^{e^{-\beta x}+1-1}}{1+e^{-\beta x}} = \beta\, \sigma(x)\left(1 - \frac{1}{1+e^{-\beta x}}\right)$$

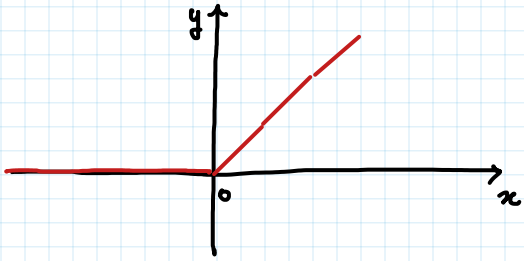
$$= \beta\, \sigma(x)\,(1-\sigma(x))$$

$$\lim_{\beta \to +\infty} \frac{1}{1+e^{-\beta x}} = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$$



2) RECTIFIED LINEAR UNIT (RELU)

$\sigma : \mathbb{R} \to \mathbb{R}$ $\quad \sigma \in \mathcal{C}^0(\mathbb{R})$ $\quad \sigma \nearrow$

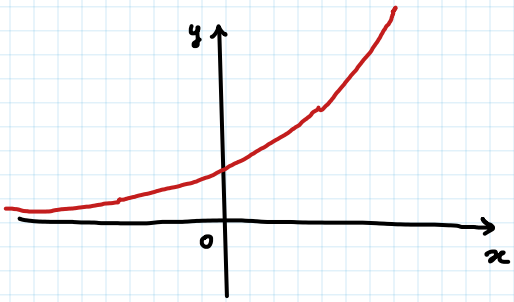


$$\sigma(x) = \max\{0, x\} = \begin{cases} 0 & \text{se } x \leq 0 \\ x & \text{altrimenti} \end{cases}$$

$$\frac{d\sigma}{dx}(x) = 0 \quad \text{se } x \leq 0 \qquad \frac{d\sigma}{dx}(x) = 1 \quad \text{se } x > 0$$

3) SOFTPLUS - SMOOTH RELU

$\sigma : \mathbb{R} \to \mathbb{R}$ $\quad \sigma \nearrow$ $\quad \sigma \in \mathcal{C}^\infty(\mathbb{R})$



$$\sigma_\beta(x) = \frac{1}{\beta} \log(1 + e^{\beta x})$$

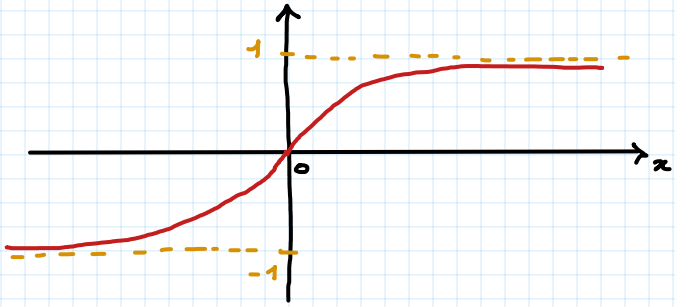
HW : $\dfrac{d\sigma}{dx} = \dots$

$\lim_{\beta \to +\infty} \sigma_\beta(x) = \dots$

4) TANGENTE IPERBOLICA – TANH

$\sigma : \mathbb{R} \to \mathbb{R}$ $\quad \sigma \nearrow$ $\quad |\sigma| \leq 1$ $\quad \sigma \in \mathcal{C}^{\infty}(\mathbb{R})$

$\beta \in \mathbb{R}$, $\beta > 0$



$$\sigma_\beta(x) = \tanh(\beta x) = \frac{e^{\beta x} - e^{-\beta x}}{e^{\beta x} + e^{-\beta x}} = \frac{e^{2\beta x} - 1}{e^{2\beta x} + 1} = \frac{1 - e^{-2\beta x}}{1 + e^{-2\beta x}}$$

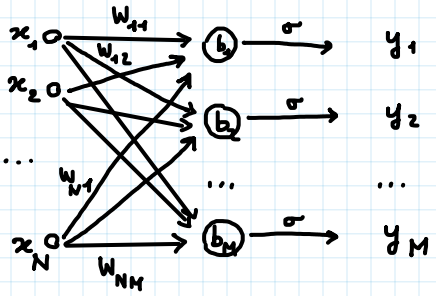
HW: $\frac{d\sigma}{dx}(x) = \dots$

$$\lim_{\beta \to +\infty} \sigma_\beta(x) = \dots = \begin{cases} 1 & \text{se } x > 0 \\ 0 & \text{se } x = 0 \\ -1 & \text{se } x < 0 \end{cases}$$

Sigmoide logistica

$$\sigma_{2\beta}(x) = \frac{1}{2}\left(\tanh_\beta(\beta x) + 1\right)$$

RETI NEURALI → SINGLE LAYER PERCEPRON



STRATO di INPUT

STRATO di OUTPUT

FULLY CONNECTED

Def: Si dice SLP un GRAFO $G = \langle V, E \rangle$ DIRETTO, BIPARTITO, ACICLICO i cui vertici sono NEURONI ARTIFICIALI

$(a,b) \neq (b,a)$

A, B si chiamano STRATI (LAYERS) $\begin{cases} A, B \subseteq V \text{ t.c. } A \cap B = \emptyset \\ \wedge\ A \cup B = V \end{cases}$

$$f : \mathbb{R}^N \to \mathbb{R}^M$$

PARAMETRI: $\begin{bmatrix} w_{11} & w_{21} & \dots & w_{N1} \\ w_{12} & w_{22} & \dots & w_{N2} \\ \dots & \vdots & & \dots \\ w_{1M} & w_{2M} & & w_{NM} \end{bmatrix}$ MATRICE di PESI $\begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_M \end{bmatrix}$ VETTORE di BIAS

$f : \mathbb{R}^N \to \mathbb{R}^M \qquad \theta = (W, \underline{b})$ $\qquad$ W matrice di PESI, $\underline{b}$ vettore di BIAS

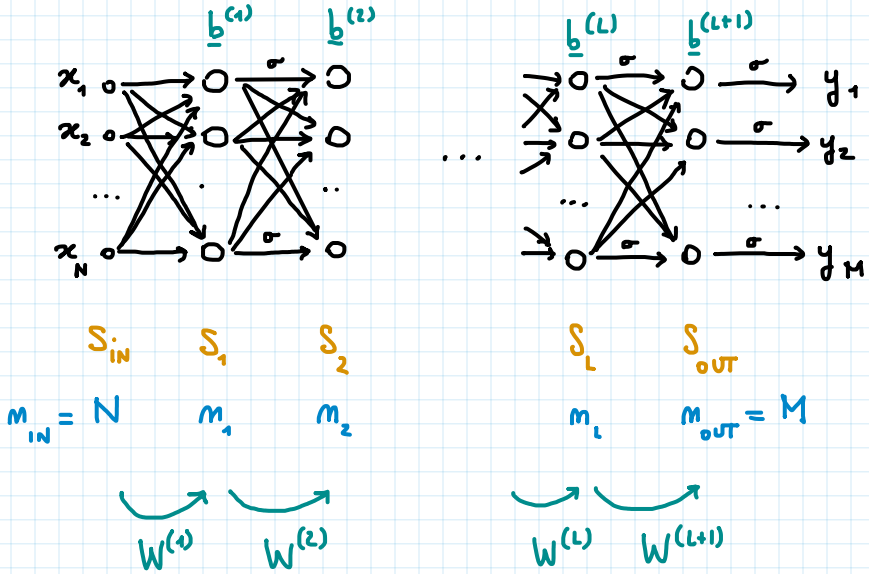
$$f_\theta(\underline{x}) = \begin{bmatrix} \sigma(\underline{x} \cdot \underline{W}_1 + b_1) \\ \sigma(\underline{x} \cdot \underline{W}_2 + b_2) \\ \vdots \\ \sigma(\underline{x} \cdot \underline{W}_M + b_M) \end{bmatrix}$$

W_i sono le COLONNE

$W \in \mathbb{R}^{N \times M}$

$$f_\theta(\underline{x}) = \left(\sigma\left(\sum_{i=1}^{N} w_{ij} \cdot x_i + b_j \right) \right)_{j=1}^{M}$$

RETI NEURALI - MULTI-LAYER PERCEPTRON



FULLY CONNECTED

$$F : \mathbb{R}^N \longrightarrow \mathbb{R}^{m_1} \longrightarrow \mathbb{R}^{m_2} \to \dots \quad \dots \mathbb{R}^L \longrightarrow \mathbb{R}^M$$

Def. Si dice MLP un GRAFO $G = \langle V, E \rangle$ DIRETTO e ACICLICO con $\underbrace{L \text{ STRATI}}_{\text{STRATI NASCOSTI}}$ + 1 strato di INPUT + 1 strato di OUTPUT :

t.c. sono a due a due DISGIUNTI e $S_{IN} \cup S_1 \cup \dots \cup S_L \cup S_{OUT} = V$

$\forall$ sottografo $G_{ij} = \langle S_i \cup S_j, E \rangle$ sono BIPARTITI.

$$\begin{cases} F_0(\underline{x}) = \underline{x} \\ F_k(\underline{x}) = \sigma\left(W^{(k)T} \cdot F_{k-1}(\underline{x}) + \underline{b}^{(k)} \right) \end{cases}$$

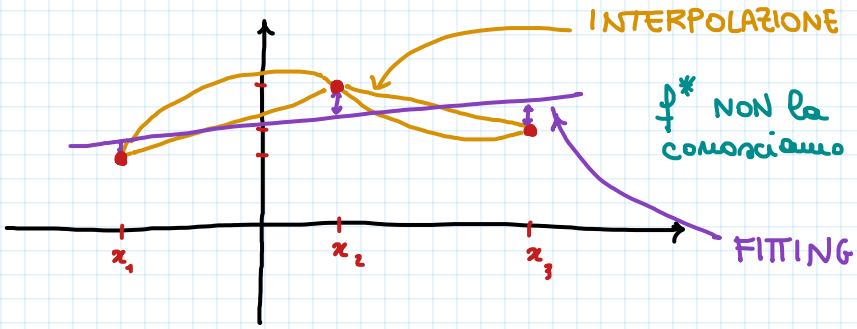
ESEMPIO di MLP con 1 STRATO NASCOSTO

$$F(\underline{x}) = \sigma\left(W^{(2)T} \cdot \boxed{\sigma\left(\underbrace{W^{(1)T} \cdot \underline{x} + \underline{b}^{(1)}}_{\circledast} \right)} + \underline{b}^{(2)} \right)$$

ATTIVAZIONE del I strato

$\circledast$ · PRE-ATTIVAZIONE $\underline{w} \cdot \underline{x} + b$

PROBLEMA : Trovare i parametri $\theta = (W, \underline{b})$ t.c. il MODELLO F_θ APPROSSIMI BENE una funzione obiettivo $f^* : \mathbb{R}^N \to \mathbb{R}^M$.

STRATEGIA : APPRENDIMENTO SUPERVISIONATO

$$D = \{ (\underline{x}, f^*(\underline{x})) : \underline{x} \in \mathbb{R}^N \}$$